

LAPORAN PRAKTIKUM 11

Enhance Industrial Software Quality Engineering and Automation Testing

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1. Pendahuluan

Software testing merupakan proses pengujian perangkat lunak untuk memastikan sistem berjalan sesuai requirement dan bebas dari defect.

2. Studi Kasus

Project Team Task Manager dengan fitur tambah task, update status, delete task, dan search task.

3. Requirement Sistem

ID	Requirement
REQ-01	User dapat menambah task
REQ-02	Task wajib memiliki title
REQ-03	Priority hanya low/medium/high
REQ-04	Status hanya todo/in_progress/done
REQ-05	User dapat menghapus task
REQ-06	User dapat mencari task
REQ-07	Sistem menolak input kosong
REQ-08	Sistem stabil terhadap invalid input

4. Risk Analysis

Fitur	Probability	Impact	Risk Level
Add Task	Tinggi	Tinggi	Sangat Tinggi
Delete Task	Sedang	Tinggi	Tinggi
Search Task	Sedang	Sedang	Sedang
Update Status	Tinggi	Sedang	Tinggi

Berdasarkan hasil analisis risiko, fitur Add Task memiliki tingkat risiko paling tinggi karena merupakan fitur utama yang sangat berpengaruh terhadap keseluruhan sistem.

5. Test Plan – Environment

Komponen	Detail
Operating System	Windows 11
Programming Language	Python 3.12

Testing Framework	Pytest
Repository	GitHub
CI/CD	GitHub Actions

6. Test Case

TC ID	Scenario	Input	Expected	Actual	Status
TC-01	Add task valid	title valid	Task tersimpan	Sesuai	Pass
TC-02	Delete task valid	task id valid	Task terhapus	Sesuai	Pass
TC-03	Search task valid	keyword valid	Task ditemukan	Sesuai	Pass
TC-04	Add task tanpa title	title kosong	Sistem menolak	Sesuai	Pass
TC-05	Priority invalid	priority = urgent	Sistem menolak	Sesuai	Pass
TC-06	Status invalid	status = finish	Sistem menolak	Sesuai	Pass
TC-07	Input title panjang	255 karakter	Sistem tetap stabil	Sesuai	Pass
TC-08	Delete task tidak ada	id invalid	Sistem menolak	Sesuai	Pass
TC-09	Search task kosong	keyword kosong	Sistem tetap berjalan	Sesuai	Pass

Edge Case Testing

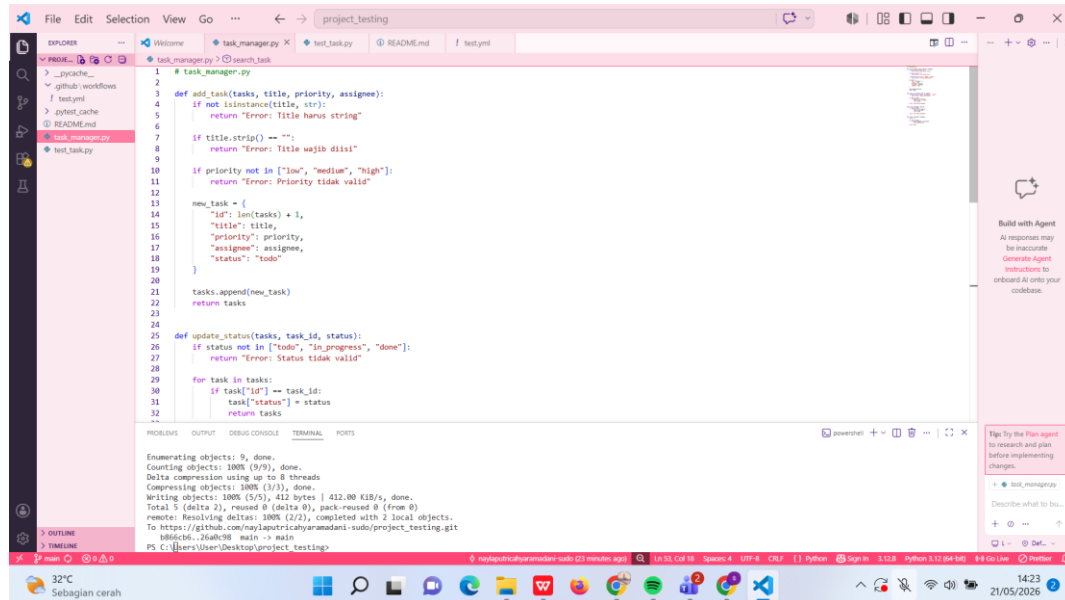
EC ID	Scenario	Expected Result	Status
EC-01	Input title sangat panjang	Sistem tetap stabil	Pass
EC-02	Search dengan keyword kosong	Sistem tidak crash	Pass
EC-03	Delete task yang tidak tersedia	Sistem menampilkan error validasi	Pass

7. Requirement Traceability Matrix (RTM)

Requirement	Test Case	Status
REQ-01	TC-01	Pass
REQ-02	TC-04	Pass
REQ-03	TC-05	Pass
REQ-04	TC-06	Pass
REQ-05	TC-02, TC-08	Pass
REQ-06	TC-03, TC-09	Pass
REQ-07	TC-04	Pass
REQ-08	TC-05, TC-06, TC-07	Pass

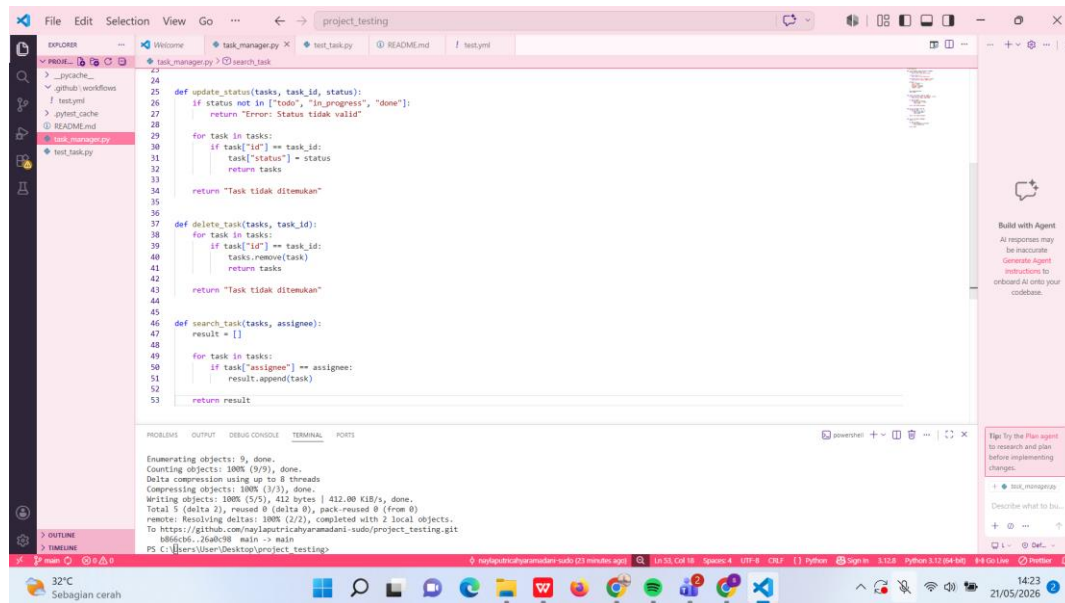
8. Source Code

a. task_manager.py



```
1 # task_manager.py
2
3 def add_task(tasks, title, priority, assignee):
4     if not isinstance(title, str):
5         return "Error: Title harus string"
6
7     if title.strip() == "":
8         return "Error: Title wajib diisi!"
9
10    if priority not in ["low", "medium", "high"]:
11        return "Error: Priority tidak valid"
12
13    new_task = {
14        "id": len(tasks) + 1,
15        "title": title,
16        "priority": priority,
17        "assignee": assignee,
18        "status": "todo"
19    }
20    tasks.append(new_task)
21    return tasks
22
23
24
25 def update_status(tasks, task_id, status):
26     if status not in ["todo", "in progress", "done"]:
27         return "Error: Status tidak valid"
28
29     for task in tasks:
30         if task["id"] == task_id:
31             task["status"] = status
32             return tasks
33
34
35
36
37 def delete_task(tasks, task_id):
38     for task in tasks:
39         if task["id"] == task_id:
40             tasks.remove(task)
41             return tasks
42
43     return "Task tidak ditemukan"
44
45
46
47 def search_task(tasks, assignee):
48     result = []
49
50     for task in tasks:
51         if task["assignee"] == assignee:
52             result.append(task)
53
54     return result
```

Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 412 bytes | 412.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/naylaputricahyanamadani-sudo/project_testing.git
bb6bcb1..26ab1b8 main -> main
PS C:\Users\User\Desktop\project_testing



```
24
25 def update_status(tasks, task_id, status):
26     if status not in ["todo", "in progress", "done"]:
27         return "Error: Status tidak valid"
28
29     for task in tasks:
30         if task["id"] == task_id:
31             task["status"] = status
32             return tasks
33
34     return "Task tidak ditemukan"
35
36
37 def delete_task(tasks, task_id):
38     for task in tasks:
39         if task["id"] == task_id:
40             tasks.remove(task)
41             return tasks
42
43     return "Task tidak ditemukan"
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46 def search_task(tasks, assignee):
47     result = []
48
49     for task in tasks:
50         if task["assignee"] == assignee:
51             result.append(task)
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53     return result
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bb6bcb1..26ab1b8 main -> main
PS C:\Users\User\Desktop\project_testing

b. test_task.py

The image shows two screenshots of a VS Code editor window. The top screenshot displays the file `test_task.py` with the following code:

```
1 from task_manager import {
2     add_task,
3     update_status,
4     delete_task,
5     search_task
6 }
7
8
9 def test_add_task_valid():
10     tasks = []
11     result = add_task(
12         tasks,
13         "Belajar Testing",
14         "high",
15         "Rina"
16     )
17
18
19     assert len(result) == 1
20
21
22 def test_add_task_empty():
23     tasks = []
24
25     result = add_task(
26         tasks,
27         "",
28         "high",
29         "Rina"
30     )
31
32     assert result == "Error: Title wajib diisi"
```

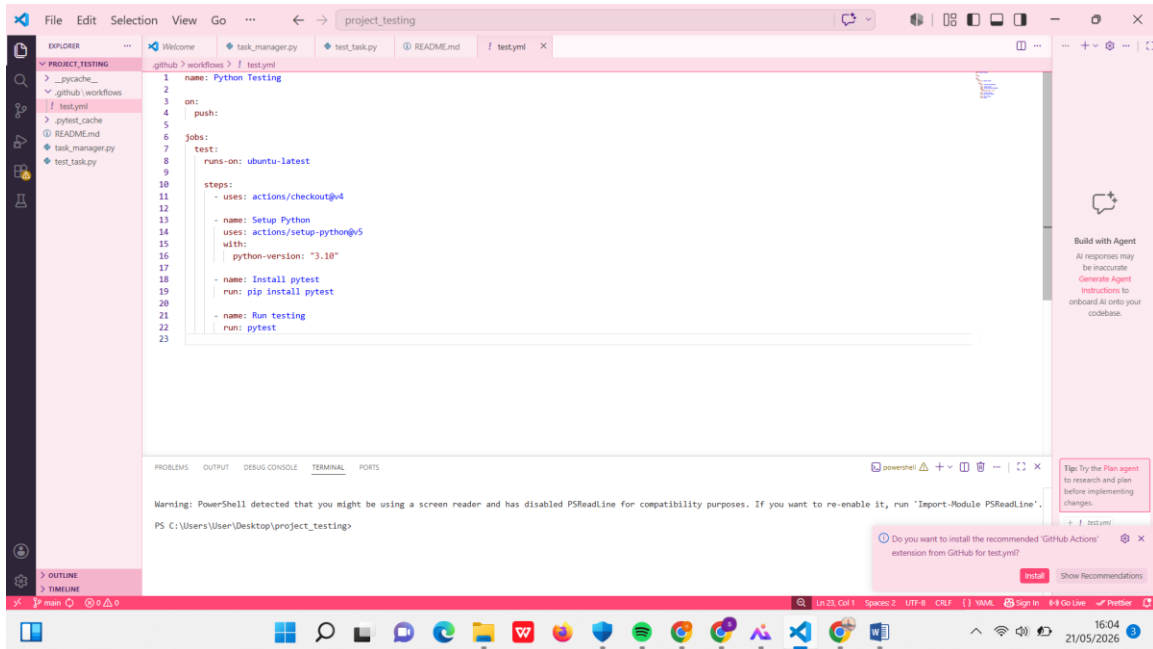
The bottom screenshot displays the file `test_update_status_invalid.py` with the following code:

```
33
34
35 def test_invalid_priority():
36     tasks = []
37
38     result = add_task(
39         tasks,
40         "Task Baru",
41         "urgent",
42         "Rina"
43     )
44
45     assert result == "Error: Priority tidak valid"
46
47
48 def test_update_status_valid():
49     tasks = [
50         {
51             "id": 1,
52             "status": "todo"
53         }
54     ]
55
56     result = update_status(
57         tasks,
58         1,
59         "done"
60     )
61
62     assert result[0]["status"] == "done"
63
64
```

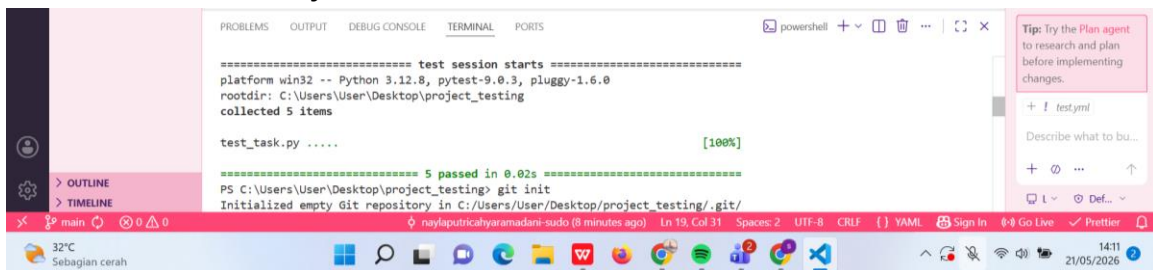
Both screenshots show the terminal output at the bottom, which includes the following text:

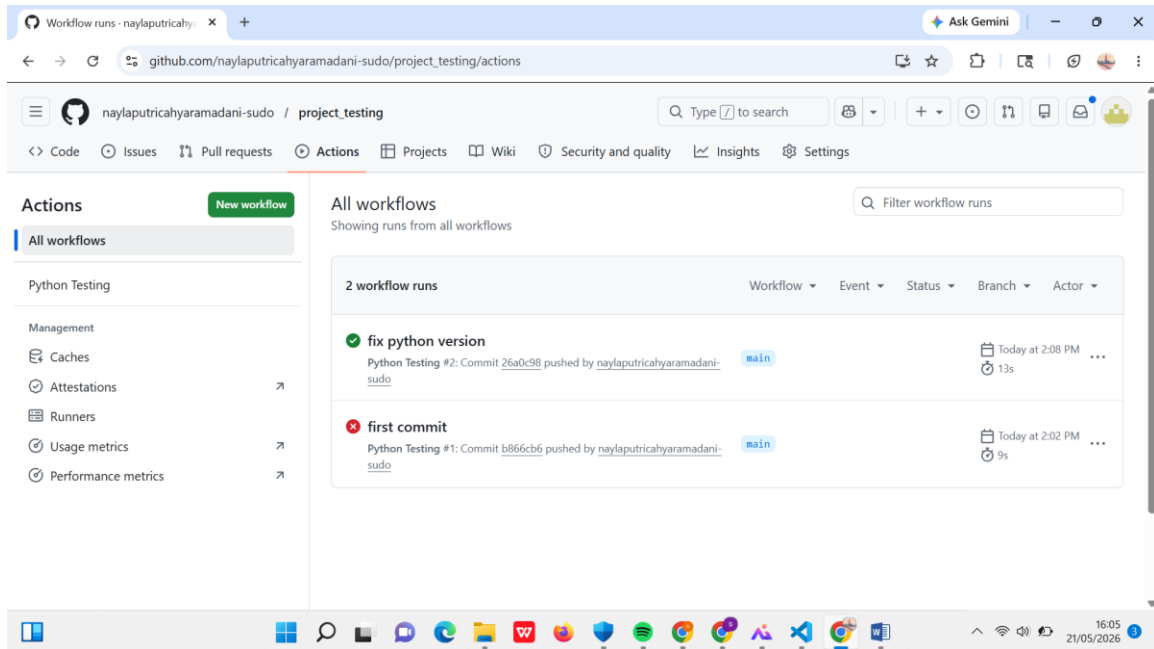
```
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 412 bytes | 412.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/nayaputriahyandani-sudo/project_testing.git
b86c46..26a4c38 main -> main
PS C:\Users\User\Desktop\project_testing>
```

c. test.yml



9. Screenshot Hasil Pytest





10. Defect Report

Field	Isi
Defect ID	BUG-01
Summary	Invalid priority diterima
Severity	Medium
Priority	High
Status	Closed

11. Analisis Metrics Testing

Metric	Nilai
Total Test Case	9
Passed	9
Failed	0
Pass Rate	100%

a. Apakah seluruh requirement sudah diuji?

Ya, seluruh requirement telah diuji menggunakan positive testing, negative testing, dan edge case testing.

b. Defect apa yang ditemukan?

Defect yang ditemukan adalah sistem menerima priority invalid pada saat proses add task.

c. Apa penyebab defect?

Defect terjadi karena belum adanya validasi input priority pada fungsi `add_task`.

d. Apa risiko jika defect lolos?

Jika defect lolos ke production, maka data task dapat menjadi tidak konsisten dan mengganggu proses manajemen task.

e. Apakah sistem siap digunakan?

Sistem sudah siap digunakan karena seluruh test case berhasil dijalankan dan defect utama telah diperbaiki.

12. Kesimpulan

Berdasarkan hasil testing menggunakan Pytest dan GitHub Actions, seluruh fitur utama Task Manager berhasil diuji dan berjalan sesuai requirement. Automation testing membantu memastikan kualitas software lebih stabil dan mengurangi risiko defect pada production.

Penggunaan GitHub Actions juga membantu proses Continuous Integration sehingga pengujian dapat berjalan otomatis setiap terdapat perubahan pada repository.

13. Link GitHub Repository

https://github.com/naylaputricahyaramadani-sudo/project_testing